

# Supplementary Material for “Real-Time AI Service Economy: A Framework for Agentic Computing Across the Continuum”

This document provides supplementary material for the main paper, including background definitions with a full notation reference, detailed proof sketches, a related work comparison, simulation parameter specifications with result summary tables, and detailed statistical analysis.

## I. BACKGROUND DEFINITIONS

The formal results in the main paper rely on three standard concepts from combinatorial optimisation and economic theory. We collect their definitions here for self-containedness; readers familiar with these concepts may proceed directly to the proofs. Notation is provided in Table I.

### A. Polymatroid

**Definition 1** (Polymatroid). *Let  $E$  be a finite ground set and  $\rho: 2^E \rightarrow \mathbb{R}_{\geq 0}$  a set function that is (i) normalised:  $\rho(\emptyset) = 0$ ; (ii) monotone:  $S \subseteq T \Rightarrow \rho(S) \leq \rho(T)$ ; and (iii) submodular:  $\rho(S \cup \{e\}) - \rho(S) \geq \rho(T \cup \{e\}) - \rho(T)$  for all  $S \subseteq T$  and  $e \notin T$ . The polymatroid associated with  $\rho$  is the polytope*

$$P(\rho) = \{x \in \mathbb{R}_{\geq 0}^E : x(S) \leq \rho(S) \text{ for all } S \subseteq E\},$$

where  $x(S) = \sum_{e \in S} x_e$ . The function  $\rho$  is called the rank function of the polymatroid [1], [2].

*Intuition.* A polymatroid captures a system of capacity constraints with diminishing returns: the more resources already committed within a subset, the less additional throughput any new resource can contribute. In the context of this paper, the ground set  $E$  corresponds to the leaf services consumed by agents, and the rank function  $\rho$  encodes the bottleneck capacities imposed by internal nodes of the service-dependency DAG. When the DAG is a tree or series-parallel network, its constraint sets form a *laminar* (nested) family, and the resulting feasible region is a polymatroid. This structure is important for two reasons: (i) Edmonds’ [1] greedy algorithm maximises any separable concave objective (including social welfare) over a polymatroid in polynomial time, and (ii) polymatroidal feasible sets are compatible with gross-substitutes valuations, jointly guaranteeing market equilibrium existence.

### B. Walrasian (Competitive) Equilibrium

**Definition 2** (Walrasian Equilibrium). *An allocation-price pair  $(x^*, p^*)$  is a Walrasian (or competitive) equilibrium if (i) every agent maximises its utility at prices  $p^*$ , and (ii) aggregate demand equals aggregate supply (market clearing).*

*Intuition.* A Walrasian equilibrium is a vector of per-resource prices at which every agent is individually satisfied

with its allocation and no resource is over- or under-demanded. Its existence implies that fully decentralised coordination via prices alone can replicate the outcome of a benevolent central planner. The Kelso–Crawford [3] fixed-point argument guarantees existence whenever agents’ valuations satisfy the gross-substitutes condition (defined below), which holds in this paper’s setting under slice encapsulation (Lemma 1 in the main paper).

### C. Gross Substitutes and Dominant-Strategy Incentive Compatibility

**Definition 3** (Gross-Substitutes (GS) Condition). *An agent’s valuation function satisfies the gross-substitutes condition if, whenever the price of one item increases (with all other prices unchanged), the agent’s demand for every other item whose price did not change weakly increases [4].*

**Definition 4** (Dominant-Strategy Incentive Compatibility (DSIC)). *A mechanism is dominant-strategy incentive compatible (DSIC) if truthful reporting of valuations maximises each agent’s utility regardless of what other agents report [5].*

*Intuition.* The GS condition captures the idea that resources are interchangeable rather than complementary: raising the price of one service slice causes an agent to substitute towards alternatives rather than to abandon the market entirely. This rules out the “I need both or neither” complementarity pattern that makes combinatorial auctions intractable. When GS holds on a polymatroidal feasible set, classical results [4], [3] guarantee the existence of a Walrasian equilibrium, and the VCG mechanism implements the welfare-maximising allocation in dominant strategies (DSIC): no agent can gain by misreporting, making the market robust to strategic behaviour. Under additional single-parameter conditions (e.g., when each task maps to a unique slice type so that each bidder’s private information is a scalar willingness-to-pay), the ascending clinching auction [6], [7] provides an alternative DSIC implementation with the added benefit of weak budget balance.

## II. PROOF DETAILS

### A. Proof of Proposition 1 (Polymatroidal Structure under Tree/SP DAGs)

**Proposition 1** (Polymatroidal Structure under Tree/SP DAGs). *Let  $G_{\text{res}}$  be a rooted tree or two-terminal series-parallel network with node capacities  $C_v > 0$ . Then:*

- (i) *the capacity function  $f$  (the rank function defined by the laminar constraint family) is submodular, monotone, and normalised ( $f(\emptyset) = 0$ );*

TABLE I  
SUMMARY OF NOTATION

| Symbol                     | Meaning                                                             |
|----------------------------|---------------------------------------------------------------------|
| $\mathcal{A}$              | Set of agents (consumers, providers, integrators, AI agents)        |
| $t$                        | Discrete time index, $t = 1, 2, \dots$                              |
| $\theta_i(t)$              | Type of agent $i$ at time $t$                                       |
| $\Theta_i$                 | Type space for agent $i$                                            |
| $\ell_i$                   | Physical location or region of agent $i$                            |
| $B_i(t)$                   | Budget or credit of agent $i$                                       |
| $\mathcal{T}_i(t)$         | Tasks issued or served by agent $i$                                 |
| $\kappa_i$                 | Capabilities of agent $i$                                           |
| $\rho_i$                   | Role of agent $i$ (consumer, provider, integrator)                  |
| $\phi_i(t)$                | Governance attributes (trust, compliance, policy flags)             |
| $\eta_k(t)$                | Parameters of task $k$ (quality, deadline, sensitivity)             |
| $q_k$                      | Quality or workload of task $k$                                     |
| $d_k^{\max}$               | Maximum acceptable latency (deadline)                               |
| $\lambda_k$                | Latency sensitivity parameter                                       |
| $T_{ik}$                   | Latency experienced by task $k$ served by agent $i$                 |
| $v_{ik}(q)$                | Base value from completing task $k$ at quality $q$                  |
| $\delta_{ik}(T)$           | Latency discount factor                                             |
| $V_{ik}(T, q)$             | Latency-aware task value for agent $i$                              |
| $\mathcal{R}$              | Set of service types                                                |
| $C_{r,\ell}(t)$            | Capacity of service $r$ at location $\ell$                          |
| $G_{\text{res}}$           | Service-dependency DAG                                              |
| $L(G)$                     | Set of leaf services in DAG $G$                                     |
| $f$                        | Polymatroid rank function of $\mathcal{X}_{\text{res}}$             |
| $s_t$                      | System state at time $t$                                            |
| $\Lambda(t)$               | Network latency conditions across the continuum                     |
| $D(t)$                     | Exogenous demand or load                                            |
| $R(t)$                     | Reputation or trust states of all agents                            |
| $r_i(t)$                   | Reputation score of agent $i$ , $r_i(t) \in [0, 1]$                 |
| $G(t)$                     | Governance/policy state at time $t$                                 |
| $x_t$                      | Allocation at time $t$                                              |
| $P_i(t)$                   | Net payment by agent $i$ ( $> 0$ : pays; $< 0$ : receives)          |
| $\mathcal{X}_t$            | Feasible allocation set                                             |
| $\mathcal{X}_{\text{res}}$ | Service- and DAG-based feasibility constraints                      |
| $\mathcal{X}_{\text{gov}}$ | Governance feasibility constraints                                  |
| $m_i(t)$                   | Message sent by agent $i$                                           |
| $\mathcal{M}_i$            | Message space for agent $i$ ( $= \Theta_i$ under direct mechanisms) |
| $\mathcal{M}$              | Mechanism mapping messages to $(x_t, P(t))$                         |
| $W(t)$                     | Social welfare at time $t$                                          |
| $\mu$                      | Externality trade-off parameter ( $\mu \geq 0$ )                    |
| $u_i(t)$                   | Per-period utility of agent $i$                                     |
| $\gamma_i$                 | Risk-aversion coefficient                                           |
| $\text{cost}_j(x_t)$       | Operational cost of provider $j$ under allocation $x_t$             |

- (ii)  $\mathcal{X}_{\text{res}}$  is a polymatroid with rank function  $f$ ; and  
 (iii) the base polytope  $B(f) = \{x \in \mathcal{X}_{\text{res}} : x(L(G)) = f(L(G))\}$  is the set of tight allocations.

*Proof.* For tree-structured DAGs, the constraint sets  $\{L_v\}$  form a laminar family by the nesting property of subtrees. Edmonds [1] established that the polytope  $\{x \geq 0 : x(L_v) \leq C_v\}$  defined by a laminar family is a polymatroid. Submodularity of  $f$  follows from the structure of laminar covers: for any  $S' \supseteq S$  and element  $l \notin S'$ , the marginal contribution  $f(S' \cup \{l\}) - f(S')$  is bounded by the capacity of the smallest constraint set containing  $l$ , which can only become a tighter bottleneck as  $S'$  grows [2]. Monotonicity is immediate from

non-negative capacities;  $f(\emptyset) = 0$  is trivial.

For series-parallel (SP) networks, we use the standard recursive definition of two-terminal SP graphs (see, e.g., [2]). An SP graph  $G = (V, E)$  with designated source  $s$  and sink  $t$  is built from primitive edges via series and parallel composition. We adopt the following structural conventions, which are maintained as invariants throughout the induction:

- (I1) *Leaf semantics.* The leaf set  $L(G)$  consists of all sink-adjacent terminal nodes that correspond to services consumed by agents. Every internal (non-leaf) node  $v$  has a well-defined leaf block  $L_v = \{l \in L(G) : l \text{ is reachable from } v\}$ .
- (I2) *Disjoint components.* Before composition, the two component graphs  $G_1, G_2$  have disjoint node sets (except for the identified glue node in series composition).
- (I3) *Glue-node semantics.* In series composition  $G = G_1 \cdot G_2$  (identifying  $t_1 = s_2$ ), the glue node  $t_1 = s_2$  becomes internal in  $G$ , and its leaf block in  $G$  is the union of the leaves reachable through  $G_2$  and any leaves reachable through  $G_1$  via the glue node. Specifically, for any node  $v$  in  $G_1$  that is an ancestor of  $t_1$ , the composed leaf block is  $L_v = L_v^{(1)} \cup L(G_2)$ , where  $L_v^{(1)} = \{l \in L(G) \cap V(G_1) : l \text{ reachable from } v\}$  is the restriction of  $v$ 's reachable leaves to those that remain leaves in  $G$  (excluding  $t_1$ , which becomes internal).

*Induction hypothesis:* for any SP graph built from fewer than  $k$  primitive edges satisfying (I1)–(I3), the family  $\{L_v : v \text{ internal}\}$  is laminar (every two members are nested or disjoint).

*Base case.* A single edge  $(s, t)$  with capacity  $C_s$ : here  $s$  is the sole internal node and  $t$  is the sole leaf, so the family  $\mathcal{L} = \{L_s\} = \{\{t\}\}$  is indexed by the single internal node  $s$  and is trivially laminar. Invariants (I1)–(I3) hold vacuously.

*Series composition.* Let  $G_1$  (source  $s_1$ , sink  $t_1$ ) and  $G_2$  (source  $s_2$ , sink  $t_2$ ) have laminar families  $\mathcal{L}_1$  and  $\mathcal{L}_2$  over disjoint leaf sets  $L_1$  and  $L_2$ , respectively (by invariant (I2)). The composed graph  $G = G_1 \cdot G_2$  identifies  $t_1 = s_2$ , with source  $s_1$  and sink  $t_2$ . Consider any two members  $L_v, L_w$  in the composed family  $\mathcal{L}$ . There are three cases: (a) both  $v, w$  lie in  $G_2$ , so  $L_v, L_w \in \mathcal{L}_2$  and laminarity follows by induction; (b)  $v$  lies in  $G_1$  and  $w$  in  $G_2$ : if  $v$  is an ancestor of  $t_1$  in  $G_1$ , then  $L_v = L_v^{(1)} \cup L_2$  where  $L_v^{(1)} \subseteq L_1$ , and  $L_w \subseteq L_2$ , so  $L_w \subset L_v$  (nested); if  $v$  is not an ancestor of  $t_1$ , then  $L_v \subseteq L_1$  and  $L_w \subseteq L_2$ , hence disjoint; (c) both  $v, w$  lie in  $G_1$ : every internal node of a two-terminal SP graph lies on a source-sink path, so all internal nodes of  $G_1$  reach  $t_1$ ; thus both  $v$  and  $w$  are ancestors of  $t_1$ . We show their composed leaf blocks are nested. The sink of any SP graph produced by this construction is always a leaf (base case:  $t$  is the sole leaf; parallel composition: the shared sink remains terminal; series composition: the new sink  $t_2$  is a leaf of  $G_2$  by induction). Since both  $v$  and  $w$  reach  $t_1$  and  $t_1 \in L(G_1)$ , the leaf blocks of  $v$  and  $w$  within  $G_1$ 's own leaf set (which includes  $t_1$ ) both contain  $t_1$ . Hence these  $G_1$ -leaf-blocks are not disjoint; laminarity of  $\mathcal{L}_1$  forces nesting, say the  $G_1$ -leaf-block of  $v$  is contained in that of  $w$ . Removing  $t_1$  preserves the inclusion:  $L_v^{(1)} \subseteq L_w^{(1)}$ . Therefore  $L_v = L_v^{(1)} \cup L_2 \subseteq L_w^{(1)} \cup L_2 = L_w$

(nested). Hence  $\mathcal{L}$  is laminar.

*Invariant preservation (series).* (I1): the leaf set of  $G$  is  $L_1 \cup L_2$ ; the glue node  $t_1 = s_2$  is internal, and every internal node  $v$  has a well-defined  $L_v$  as constructed above. (I2): the only shared node is the glue node  $t_1 = s_2$ ; all other nodes are disjoint by induction. (I3): for any ancestor  $v$  of the glue node in  $G_1$ , the composed leaf block  $L_v = L_v^{(1)} \cup L(G_2)$  is exactly the definition in (I3). For nodes in  $G_2$  or non-ancestor nodes in  $G_1$ , leaf blocks are unchanged. Hence (I1)–(I3) are preserved.

*Parallel composition.* Let  $G_1$  and  $G_2$  share source  $s$  and sink  $t$ , with disjoint internal nodes and laminar families  $\mathcal{L}_1, \mathcal{L}_2$  over disjoint leaf sets  $L_1, L_2$  (by invariant (I2)). The composed family is  $\mathcal{L} = \mathcal{L}_1 \cup \mathcal{L}_2 \cup \{L_1 \cup L_2\}$ , where  $L_1 \cup L_2 = L_s$  is the leaf set of source  $s$ . Members within  $\mathcal{L}_1$  (resp.  $\mathcal{L}_2$ ) are laminar by induction; any  $L_v \in \mathcal{L}_1$  and  $L_w \in \mathcal{L}_2$  satisfy  $L_v \subseteq L_1$  and  $L_w \subseteq L_2$ , hence disjoint; and  $L_s$  contains all members as subsets. Hence  $\mathcal{L}$  is laminar.

*Invariant preservation (parallel).* (I1): the leaf set of  $G$  is  $L_1 \cup L_2$ ; the shared source  $s$  has leaf block  $L_s = L_1 \cup L_2$ , and all other internal nodes retain their original leaf blocks. (I2):  $G_1$  and  $G_2$  share only  $s$  and  $t$ ; internal nodes are disjoint by induction. (I3): no glue-node identification occurs (the shared source and sink are already present in both components), so (I3) applies vacuously. Hence (I1)–(I3) are preserved.

Since  $\mathcal{L}$  is laminar in both cases, the rank function  $f$  is submodular by the standard laminar-family result [1], [2], and the composed feasible region is a polymatroid. Parts (ii) and (iii) are standard consequences of the polymatroid definition [2].  $\square$

## B. Proof of Lemma 1 (GS Valuations under Slice Encapsulation)

**Lemma 1** (GS Valuations under Slice Encapsulation). *Suppose integrators encapsulate multi-resource service paths into composite slices, where each slice  $s$  is a discrete, indivisible unit with fixed internal routing, deterministic latency  $T_s$ , and quality  $q_s$ . Suppose further that: (a) each agent requires at most one slice per task (unit demand); (b) an agent's tasks are additively separable—its valuation decomposes as a sum across tasks with no cross-task complementarity or shared feasibility coupling; and (c) within each mechanism epoch, slice attributes  $(T_s, q_s)$  are determined by the current system state  $s_t$  and remain fixed throughout the allocation round. Then the agent's induced valuation over slices satisfies the gross-substitutes condition.*

*Proof.* Under encapsulation, agent  $i$ 's valuation for assigning task  $k$  to slice  $s$  is  $V_{ik}(T_s, q_s)$ , where  $(T_s, q_s)$  are fixed within the current epoch by condition (c). Since each task requires exactly one slice, the agent selects whichever slice maximises  $V_{ik}(T_s, q_s) - p_s$ , where  $p_s$  is the slice price. This defines a unit-demand valuation. By Proposition 2 of Gul and Stacchetti [4], unit-demand valuations satisfy GS: raising the price of one slice causes the agent either to retain it or to switch to another, but never to drop demand for an unrelated slice whose price has not changed. For agents with multiple tasks, additive

separability (condition (b)) ensures the aggregate valuation is a finite sum of GS (specifically, unit-demand) valuations. Since finite sums of GS valuations are GS [4], additive task separability preserves the GS property. Note that shared agent-side budgets, per-agent latency coupling, or shared token caps across tasks would generally break this additivity and can break GS. Note that slice capacities (the number of available units of each type) are determined by the polymatroid  $\mathcal{X}_{\text{res}}$ ; this is a feasibility condition relevant to Proposition 2 rather than a requirement for GS itself.  $\square$

## C. Proof Sketch of Proposition 2 (Efficient Mechanism Design under Structured DAGs)

*Note.* The following proof sketch reduces each part to standard results from auction theory and discrete convex analysis; the underlying theorems are cited explicitly at each step rather than re-derived.

**Proposition 2** (Efficient Mechanism Design under Structured DAGs). *Suppose  $\mathcal{X}_t$  is polymatroidal, agents' valuations satisfy the GS condition under slice encapsulation, agents have quasilinear utility (value minus payment), and slices are discrete items with integral capacities. Then:*

- (i) *a Walrasian equilibrium exists in the discrete market of  $K$  heterogeneous slice types whose integer supply capacities are jointly governed by the polymatroidal feasibility region  $\mathcal{X}_t$  (encoding both per-type upper bounds and cross-type bottleneck constraints from shared DAG resources), with per-type prices supporting the welfare-maximising allocation [3], [4], [8], [9];*
- (ii) *under a bounded integer encoding—explicit per-slice valuations in  $\{0, \dots, V_{\max}\}$ ,  $K$  slice types,  $n$  agents, at most  $T_{\max}$  tasks per agent, and a fixed price-increment rule—the welfare-maximising allocation is computable in time polynomial in  $n, K, T_{\max}$ , and  $V_{\max}$  via a Kelso–Crawford-style ascending auction with explicit demand computation [3]; and*
- (iii) *the efficient allocation is implementable through a DSIC, individually rational mechanism: VCG [5] applies in full generality (but may run a deficit). In the restricted case where each task class maps to a unique admissible slice type—so that each agent–task pair demands at most one unit of a specific slice type and its private information reduces to a scalar willingness-to-pay for that type—the polymatroid clinching auction of Goel et al. [7], extending Ausubel [6], provides an alternative DSIC implementation that additionally guarantees weak budget balance under nonnegative valuations. When tasks may choose among multiple heterogeneous slice types, VCG remains the appropriate DSIC mechanism.*

*Proof.* Part (i): Under slice encapsulation (Lemma 1), each slice type  $s$  has an integral capacity  $c_s$  (determined by the polymatroid rank function), and agents treat each unit of each slice type as an indivisible item. Each agent has unit demand per task (one slice per task). The supply of  $K$  slice types is governed by the integer polymatroid  $P_{\mathbb{Z}}(f) = \{x \in \mathbb{Z}_{\geq 0}^K : x(S) \leq f(S) \text{ for all } S\}$ , where  $f$  is the rank

function from Proposition 1. This imposes not only per-type upper bounds  $x_s \leq f(\{s\})$  but also cross-type bottleneck constraints  $x(S) \leq f(S)$  for subsets  $S$  of types sharing internal DAG resources (integral node capacities yield an integral rank function, ensuring integer vertices [2]). With per-type prices  $p_s$  and quasilinear payoffs  $V_{ik}(T_s, q_s) - p_s$ , unit-demand valuations satisfy the GS condition [4]. Fujishige and Yang [8] established that GS valuations are equivalent to  $M^\natural$ -concave functions in the framework of discrete convex analysis; the integer polymatroid supply set is  $M$ -convex [9]. Under this equivalence, the Kelso–Crawford fixed-point construction [3] extends to polymatroid-coupled supply (see Murota [9, Ch. 11]), guaranteeing a Walrasian equilibrium with per-type prices supporting the welfare-maximising integer allocation.

Part (ii): Under GS valuations, the Walrasian equilibrium of Part (i) is welfare-maximising [4]. Since the allocation problem involves *discrete* indivisible slices with integral capacities, the appropriate algorithmic framework is the ascending auction for GS valuations. We assume the following representation: each agent  $i$  reports an explicit scalar value  $V_{ik}(T_s, q_s) \in \{0, 1, \dots, V_{\max}\}$  for each task–slice pair, where  $V_{\max}$  is bounded; the demand oracle for agent  $i$  at prices  $p$  is computable in  $O(K)$  time by maximizing  $(V_{ik} - p_s)$  over slice types for each task. Under this explicit representation, the ascending auction of Kelso and Crawford [3] terminates in a number of price-increment steps polynomial in  $n$  (agents),  $K$  (slice types),  $T_{\max}$  (maximum tasks per agent), and  $V_{\max}$  (the valuation range), since prices are bounded by  $V_{\max}$  and each step requires  $O(n \cdot T_{\max} \cdot K)$  demand-oracle work (each agent evaluates  $K$  types for each of its tasks). The precise step count depends on the price-update rule, tie-breaking conventions, and price granularity, which we do not fully specify here; what matters for our purposes is that running time is polynomial under bounded integer valuations with explicit demand oracles. For practical deployments,  $V_{\max}$  and  $K$  are small, so computational cost is modest. (For a single-agent separable concave objective over a continuous polymatroid, Edmonds’ greedy algorithm [1] provides an alternative  $O(|\mathcal{R}| \log |\mathcal{R}|)$ -time solution [2], but the multi-agent discrete-item setting of this paper relies on the ascending auction route.)

Part (iii): Within the static-epoch setting of Parts (i)–(ii), given the welfare-maximising allocation and quasilinear utility, VCG payments implement this outcome in dominant strategies (DSIC) and guarantee individual rationality (IR) [5]. (This is a per-epoch DSIC result, not a dynamic repeated-game guarantee.) We note that, in general, pairing a Walrasian allocation with VCG payments need not be DSIC, because a Walrasian allocation need not maximise social welfare. Our DSIC claim relies on the GS structure established in Lemma 1: under gross substitutes, the welfare-maximising allocation coincides with a Walrasian equilibrium allocation [4], so VCG payments apply to the exact social-welfare maximiser and the standard DSIC guarantee follows. However, VCG does not generically guarantee weak budget balance: the sum of VCG payments can be negative (a deficit) in multi-item settings.

For the clinching-auction claim, we require an explicit reduction to the single-parameter polymatroid clinching frame-

work of Goel et al. [7]. This reduction is valid when each task class maps to a *unique* admissible slice type. Under this condition, task  $k$  of agent  $i$  demands at most one unit of its designated slice type  $s(k)$ , and its private information reduces to the scalar willingness-to-pay  $V_{ik}(T_{s(k)}, q_{s(k)})$ —a single-parameter value. Additive separability across tasks (Lemma 1, condition (b)) ensures that the agent’s aggregate demand decomposes into independent per-task single-parameter demands with no cross-task coupling. The resulting environment—a collection of single-parameter bidders, each seeking one unit of their designated type, subject to a polymatroidal supply constraint—matches the polymatroidal clinching framework of Goel et al. [7], extending the ascending clinching auction of Ausubel [6]. Under this reduction, the clinching auction provides a DSIC implementation that additionally satisfies weak budget balance (non-negative revenue) under nonnegative valuations, and reveals equilibrium prices iteratively. Our setting specialises the Goel et al. framework [7] to agents with quasilinear utility and no binding budget constraints; the budget-constrained variant additionally handles agents with hard budgets, but that generalisation is not needed here since our utility model (Section II-F in the main paper) is quasilinear throughout.

The single-type-per-task condition holds naturally in the proposed architecture: Proposition 3 assumes each integrator exports a single homogeneous slice type, and when each task class is served by exactly one integrator, each task’s demand is one-dimensional. When tasks may choose among *multiple heterogeneous* slice types (different  $(T_s, q_s)$ ), a scalar willingness-to-pay  $\max_s V_{ik}(T_s, q_s)$  does not faithfully encode item-specific preferences—the mechanism may allocate a non-utility-maximising slice, breaking the single-parameter equivalence. In this general heterogeneous case, VCG remains the appropriate DSIC mechanism; extending budget-balanced DSIC implementation to heterogeneous GS valuations under polymatroid constraints is an open problem left to future work.  $\square$

#### D. Proof of Proposition 3 (Polymatroidal Encapsulation of Arbitrary DAGs)

**Proposition 3** (Polymatroidal Encapsulation of Arbitrary DAGs). *Let  $G_{\text{res}}$  be an arbitrary service-dependency DAG. Suppose a collection of integrators partitions the non-leaf nodes of  $G_{\text{res}}$  into disjoint clusters, where each cluster  $j$  manages a connected sub-DAG internally and exposes a single, homogeneous composite service (slice) with scalar capacity  $\bar{C}_j$  equal to the maximum throughput of its sub-DAG. Let  $G'$  denote the quotient graph obtained by contracting each cluster to a single node. If  $G'$  has tree or series-parallel structure, then the agent-facing feasible region is polymatroidal.*

*Proof.* Each integrator  $j$  replaces an internal sub-DAG with a single node whose capacity  $\bar{C}_j$  bounds the aggregate throughput available to the slices it exposes. Concretely,  $\bar{C}_j$  is the maximum flow from the sources to the sinks of the sub-DAG managed by integrator  $j$ . When a sub-DAG has multiple entry or exit points, a standard super-source/super-sink construction (adding a virtual source connected to all entry nodes and a

virtual sink connected to all exit nodes, each with infinite-capacity arcs) reduces the problem to a single-source single-sink max-flow. When the sub-DAG has node capacities, the standard node-splitting reduction—replacing each node  $v$  with two nodes  $v^{\text{in}}, v^{\text{out}}$  connected by an arc of capacity  $C_v$ —converts to an edge-capacity network; max-flow is then a standard polynomial-time computation (e.g., Ford–Fulkerson or push-relabel) that the integrator can recompute periodically as internal conditions change.

Every node  $v$  in the quotient graph  $G'$  carries a well-defined capacity: for contracted integrator nodes, the capacity is  $\bar{C}_v = \bar{C}_j$  (the max-flow of integrator  $j$ 's sub-DAG); for uncontracted nodes that were not absorbed into any integrator cluster, the capacity is the original  $\bar{C}_v = C_v$ . The agent-facing constraints are  $x(L'_v) \leq \bar{C}_v$  for each node  $v$  in  $G'$ , where  $L'_v$  is the set of slices in  $v$ 's sub-DAG. Since  $G'$  is tree or series-parallel by assumption, the constraint family  $\{L'_v\}$  is laminar and Proposition 1 applies directly. The exported throughput token is a single commodity, measured in the same units (e.g., requests per second) as the agent-facing allocation variable  $x_i$ ; this unit consistency is ensured by the max-flow computation, which yields a scalar in the same throughput units as the original node capacities. Internal optimisation within each integrator (scheduling, routing, and capacity allocation across the sub-DAG) is a separate, lower-level problem solved by platform-specific orchestration. Provided the scalar-summary and no-cross-coupling assumptions hold (single homogeneous exported slice per integrator, no shared leaves across integrators), the polymatroidal guarantee holds at the agent-facing interface for arbitrary internal sub-DAG topology.

*Remark (scope of scalar-summary assumption).* The scalar summary  $\bar{C}_j$  is valid when each integrator exports a single, homogeneous slice type so that feasible throughput does not depend on the mix of external requests. In realistic service graphs, feasible throughput may depend on request mix, route contention, and internal compatibility constraints; a single scalar is then insufficient. When an integrator exports multiple service types,  $\bar{C}_j$  generalises to a vector-valued capacity, and the quotient-graph polymatroid argument requires each integrator's external feasibility set to itself be polymatroidal—a stronger structural condition that we leave to future work. Additionally, if leaf nodes are shared across integrators (e.g., common cloud GPUs used by multiple domains), external coupling can re-enter through leaf capacities unless explicitly captured in the quotient-graph structure.  $\square$

### III. RELATED WORK COMPARISON

Table II positions this article relative to prior work across five research areas, identifying the specific gaps each area leaves and how the present framework addresses them.

### IV. SIMULATION DETAILS

#### A. Baseline Parameter Configuration

All simulations share a common baseline configuration. We simulate 50 autonomous agents over 200 rounds with independent Poisson task arrivals ( $\lambda = 1.0$  tasks/round/agent). Task values are drawn from Uniform $[1, 2]$  with exponential latency

discount  $\delta(T) = e^{-\lambda_l T}$ ,  $\lambda_l = 0.005$  per ms (i.e., 5.0 per second, so value decays to  $e^{-1} \approx 0.37$  at the longest deadline), and class-dependent deadlines in  $\{100, 150, 200\}$  ms. The environment spans three tiers (device, edge, cloud) with capacities  $\{200, 300, 500\}$  units; base propagation delays are  $\{5, 15, 50\}$  ms. Two latency models are used: agents form bids using a power-law congestion estimate  $\hat{L} = L_{\text{base}} + \alpha \rho^p$  ( $\alpha = 200, p = 1.2$ ), while actual execution latency follows an M/M/1-inspired queueing model  $L = L_{\text{base}} + \min(\lambda_{\text{load}} \cdot \frac{\rho}{1-\rho}, 2, 500)$  ms per tier, where  $\rho = \min(0.99, \text{demand/capacity})$  and the 500 ms cap prevents divergence near saturation. Per-task end-to-end latency is drawn from  $\mathcal{N}(L_{\text{crit}}, 0.1 L_{\text{crit}})$ , where  $L_{\text{crit}}$  is the critical-path latency through the DAG (coefficient of variation 10%). Trust scores initialise at 0.8 and update asymmetrically (+0.03 per successful SLA event, −0.08 per violation), reflecting a failure-dominant reputation model; governance violations result in hard rejection. Results are averaged over 10 random seeds for reproducibility.

Each DAG topology induces a distinct per-tier resource demand profile, reflecting the structural properties formalised in the main text. Linear and tree topologies distribute demand roughly evenly across tiers, whereas series-parallel (SP) topologies concentrate demand on cloud resources (reflecting parallel inference branches), and entangled topologies create heavy, asymmetric demand on device tiers with cross-tier coupling. These demand profiles determine how rapidly per-tier capacity saturates and whether the resulting excess-demand signal induces stable or oscillatory price dynamics in the tâtonnement.

The tâtonnement runs 15 iterations per round with price step  $\eta = 0.25$  (normalised by tier capacity), a price floor of 0 and cap of 1000. The online success model uses a logistic sigmoid  $p(\text{success} | \hat{\rho}) = \sigma(a + b \hat{\rho})$  initialised at  $a = 2.0, b = -2.0$  and updated via stochastic gradient descent with learning rate 0.3. Welfare is computed as realised latency-aware value minus a congestion penalty  $\gamma_c \sum_r [\max(\rho_r - 1, 0)]^2$  with  $\gamma_c = 0.05$ . Tasks that miss their deadline receive zero value (salvage fraction = 0). In Experiment 4, the hybrid integrator uses EMA smoothing coefficient  $\beta_{\text{ema}} = 0.8$ , a 10% markup over unit resource cost, slice price step 0.15, and topology-dependent efficiency factors (0.75 for SP, 0.85 for entangled) that reduce the effective per-task resource footprint during execution. An ablation condition (“hybrid EMA only”) applies the same EMA smoothing but sets efficiency = 1.0, isolating the price-smoothing effect from the demand-reduction effect (Section IV-D). Table III consolidates all baseline parameters with their values and justifications.

#### B. Experiment-Specific Parameter Settings

Table IV summarises the factors varied in each experiment and the parameters held fixed.

#### C. Result Summary Tables

Tables V to VIII and X consolidate the key numeric results from Experiments 1–5. Values are reported as mean [bootstrap 95% CI] over 10 random seeds; latency is in milliseconds. Point estimates without CIs are shown where the bootstrap interval is negligibly narrow (width < 1 unit).

TABLE II  
COMPARISON OF RELATED WORK AND POSITIONING OF THIS ARTICLE

| Category                                                              | Strengths                                                                                                | Limitations                                                                                                                                                                         | How This Article Addresses the Gap                                                                                                                                                   |
|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Edge/Fog/Cloud Service Mgmt</b> [10], [11], [12], [13], [14], [15] | Low-latency orchestration; mobility- and multi-resource-aware management.                                | Assume centralized control; Limited modeling of multi-stage service dependencies; absence of explicit agent-driven decisions; and restricted adaptability under dynamic conditions. | Introduce an agentic layer; unify resource management with autonomous task generation and negotiation; incorporate service-DAG constraints.                                          |
| <b>Service Function Chaining</b> [16], [17], [18]                     | Optimize dependency-aware service caching; capture placement constraints.                                | Do not consider strategic agents; lack integrated economic or governance constraints; ignore market or negotiation dynamics.                                                        | Model service-dependency DAGs formally; analyse structural regimes (tree/SP) enabling stable coordination; embed into an economic/management framework.                              |
| <b>Agentic/Autonomous Systems</b> [19], [20], [21]                    | Identify autonomous AI agents as active system participants; highlight emergent coordination challenges. | No resource-DAG model; no governance-aware allocation; no management-plane integration or tractable orchestration framework.                                                        | Provide a full agent-service-resource-governance model; link agent behavior to system feasibility and market outcomes.                                                               |
| <b>Governance, Trust, Norms</b> [22], [23], [24]                      | Formalize policies, trust, locality, compliance; mature models of norm enforcement.                      | Not integrated with latency/QoS constraints, dependency-aware placement, or market-based coordination.                                                                              | Embed governance directly into feasibility ( $\mathcal{X}_{\text{gov}}$ ); model reputation-dependent access; evaluate governance-performance trade-offs.                            |
| <b>Market-Based Resource Allocation</b> [25], [26], [27], [28]        | Provide incentive-compatible mechanisms; efficient pricing and allocation for substitutable resources.   | Assume independent or substitutable items; break down under interdependent resource bundles or service graphs; no governance.                                                       | Identify structural conditions where market stability is preserved (polymatroid regimes); propose hybrid slice-based architecture insulating the market from deep complementarities. |

#### D. Experiment 4 Ablation: EMA Smoothing vs. Efficiency Factor

The hybrid architecture applies two simultaneous changes relative to the naïve baseline: (i) EMA price smoothing ( $\beta_{\text{ema}} = 0.8$ , 10% markup), and (ii) an efficiency factor (SP: 0.75, entangled: 0.85) that reduces per-task resource consumption during execution. To disentangle these effects, we run a *hybrid (EMA only)* condition with efficiency = 1.0 (no demand reduction), yielding  $3 \times 2 \times 2 \times 4 \times 10 = 480$  runs total. Table IX reports the full results.

*Price volatility.*: EMA smoothing alone accounts for the majority of the volatility reduction. For SP DAGs, the EMA-only condition achieves  $\sigma \leq 0.097$  across all agent counts and loads, matching the full hybrid and representing up to a 72% reduction from naïve levels (e.g., SP/high/ $N = 60$ : naïve  $\sigma = 0.337$ , EMA only  $\sigma = 0.097$ , full hybrid  $\sigma = 0.096$ ). For entangled DAGs, EMA alone achieves comparable volatility reduction in most conditions, with two exceptions: entangled/high at  $N = 20$  ( $\sigma = 0.141$  vs. full hybrid 0.096) and entangled/medium at  $N = 40$  ( $\sigma = 0.262$  vs. full hybrid 0.096). In these cases, the efficiency factor contributes additional stabilisation by reducing effective demand below the congestion threshold.

*Latency and welfare.*: Unlike price volatility, latency and welfare improvements depend substantially on the efficiency factor. Under SP/high/ $N = 60$ , EMA only achieves 315 ms median latency versus 181 ms for full hybrid and 197 ms for naïve: the efficiency factor's demand reduction delays the onset of queueing saturation, producing a 42% latency improvement beyond EMA alone. Similarly, welfare at SP/high/ $N = 40$  is 14.4 under EMA only (identical to naïve) versus 14.9 under full hybrid. At higher  $N$  values where the

system saturates regardless, all three conditions converge to near-total drop rates and zero welfare.

*Interpretation.*: The ablation confirms that the two components of the hybrid architecture serve distinct functions: EMA smoothing is the primary mechanism for price stabilisation (the *architectural* effect), while the efficiency factor provides latency and welfare improvements by reducing effective resource consumption (the *operational* effect). In conditions far below or far above capacity, the effects are negligible; the benefits concentrate in the congested-but-not-saturated regime.

## V. DETAILED EXPERIMENT RESULTS

This section presents figures and extended discussion for each experiment. The main paper reports condensed ablation results; here we provide the per-experiment visualisations and condition-by-condition analysis.

### A. Experiment 1: Structural Ablation

Fig. 1 summarises the results across four panels. Price volatility, measured as the standard deviation of log-returns of per-tier prices across rounds, is the key metric linking the simulation to the theoretical prediction of Proposition 1: topologies with polymatroidal feasibility regions should exhibit convergent, stable prices.

At low load ( $\lambda = 0.5$ ), all four topologies achieve comparable median latencies (126–143 ms), moderate drop rates (34–44%), and zero price volatility. Under high load ( $\lambda = 1.5$ ), linear and tree topologies maintain median latencies of 134 and 142 ms with moderate drop rates (37% and 49%) and, crucially, zero price volatility.

In contrast, the entangled DAG under high load reaches a median latency of 404 ms ( $p_{95} = 473$  ms), a drop rate of

TABLE III  
BASELINE SIMULATION PARAMETERS

| Parameter                              | Symbol               | Value                              | Justification                                          |
|----------------------------------------|----------------------|------------------------------------|--------------------------------------------------------|
| <i>General</i>                         |                      |                                    |                                                        |
| Agent count                            | $N$                  | 50                                 | Sufficient for stable averages; varied in Exp 2, 4     |
| Simulation rounds                      | —                    | 200                                | Allows tâtonnement and learning convergence            |
| Monte Carlo seeds                      | —                    | 10                                 | Standard for variance estimation                       |
| <i>Task model</i>                      |                      |                                    |                                                        |
| Arrival rate                           | $\lambda$            | 1.0 tasks/round/agent              | Poisson; varied in Exp 1 (0.5, 1.0, 1.5)               |
| Task value                             | $v$                  | Uniform[1, 2]                      | Normalised; focuses analysis on structural effects     |
| Latency decay rate                     | $\lambda_l$          | 0.005 /ms                          | Value halves at $\approx 139$ ms; spans deadline range |
| Deadlines                              | $D$                  | {100, 150, 200} ms                 | Tight to relaxed relative to tier latencies            |
| Salvage fraction                       | —                    | 0                                  | Hard deadline: missed $\Rightarrow$ zero value         |
| <i>Environment</i>                     |                      |                                    |                                                        |
| Tier capacities                        | $C_r$                | {200, 300, 500}                    | Device < edge < cloud capacity                         |
| Base propagation delays                | $L_{\text{base}}$    | {5, 15, 50} ms                     | Device < edge < cloud latency                          |
| Latency CV                             | —                    | 10%                                | Gaussian noise around critical-path latency            |
| <i>Bidding and execution models</i>    |                      |                                    |                                                        |
| Bidding congestion                     | $\hat{L}$            | $L_{\text{base}} + 200 \rho^{1.2}$ | Power-law; intentional mismatch with execution         |
| Execution latency                      | $L$                  | M/M/1 with 500 ms cap              | Standard queueing; cap prevents divergence             |
| <i>Trust</i>                           |                      |                                    |                                                        |
| Trust initialisation                   | $\tau_0$             | 0.8                                | Moderate prior trust                                   |
| Trust update (success)                 | —                    | +0.03                              | Asymmetric per [23]                                    |
| Trust update (violation)               | —                    | −0.08                              | Failure-dominant per [23]                              |
| <i>Tâtonnement and learning</i>        |                      |                                    |                                                        |
| Iterations per round                   | —                    | 15                                 | Sufficient for price convergence                       |
| Price step                             | $\eta$               | 0.25                               | Normalised by tier capacity                            |
| Price floor / cap                      | —                    | 0 / 1000                           | Prevents negative or runaway prices                    |
| Online success model                   | —                    | $\sigma(2.0 - 2.0 \hat{\rho})$     | Conservative initial prior; learns from outcomes       |
| Learning rate                          | —                    | 0.3                                | Balances responsiveness and stability                  |
| Congestion penalty                     | $\gamma_c$           | 0.05                               | Mild welfare penalty for over-utilisation              |
| <i>Hybrid integrator (Exp 4 and 5)</i> |                      |                                    |                                                        |
| EMA smoothing                          | $\beta_{\text{ema}}$ | 0.8                                | Smooths round-to-round price oscillations              |
| Markup                                 | —                    | 10%                                | Covers integrator overhead                             |
| Slice price step                       | —                    | 0.15                               | Integrator price adjustment rate                       |
| Efficiency factor (SP)                 | $\eta_{\text{SP}}$   | 0.75                               | Moderate internal scheduling gain                      |
| Efficiency factor (entangled)          | $\eta_{\text{ent}}$  | 0.85                               | Higher gain for more complex sub-DAGs                  |

99.8%, and a price volatility of  $\sigma = 0.273$ . The SP topology exhibits intermediate behaviour: median latency of 146 ms at high load but a drop rate of 89.8% and price volatility of  $\sigma = 0.112$ . Under medium load, the entangled topology already shows substantial price volatility ( $\sigma = 0.373$ ), while SP, tree, and linear remain at zero.

The utilisation panel reveals the structural mechanism: entangled DAGs push effective demand to  $1.11\times$  capacity at high load, creating persistent excess demand that drives price oscillation; SP reaches  $0.83\times$ ; tree and linear remain below the critical threshold. The SP topology is fully polymatroidal, yet exhibits non-zero volatility at high load because its demand profile pushes per-tier utilisation toward saturation, where the tâtonnement fails to converge despite equilibrium existence. The ordering (linear  $\approx$  tree < SP < entangled) reflects both structural complexity and demand-profile effects, motivating the encapsulation approach.

### B. Experiment 3: Governance Ablation

Three governance regimes are compared: no governance (all tasks share a single resource pool); moderate governance (capacity split 50/50 between compliant and general pools); and strict governance (70/30 split with trust gate  $\geq 0.75$ ).

Fig. 2 reports median latency, drop rate, service coverage, and price volatility. By construction, all allocated tasks are policy-compliant; the meaningful trade-off appears in service coverage. For the tree topology under high load, service coverage falls from 59% (no governance) to 30% (strict), while median latency remains largely unchanged (138–142 ms). For the entangled topology under high load, service coverage falls from 0.4% to 0.2%, but median latency decreases substantially from 404 ms to 266 ms (a 34% reduction), as governance excludes the most congestion-prone allocations.

The price-volatility panel reveals a secondary effect: strict governance substantially increases price volatility across all conditions, because capacity partitioning creates smaller sub-pools that saturate more easily. This is most pronounced for

TABLE IV  
EXPERIMENT-SPECIFIC PARAMETER SETTINGS

| Exp. | Varied Aspect                                | Settings                                                                                                                                  | Fixed                                                                                                                     |
|------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 1    | Service-dependency graph shape $\times$ load | Linear, tree, series-parallel (SP), entangled DAG; low, medium, high ( $\lambda \in \{0.5, 1.0, 1.5\}$ )                                  | $N = 50$ ; governance: baseline; architecture: naïve (no slicing)                                                         |
| 2    | Agent population $\times$ topology           | $N \in \{10, 20, 30, 40, 50, 60\}$ ; tree, SP, and entangled topologies                                                                   | Load: medium ( $\lambda = 1.0$ ); governance: baseline; architecture: naïve                                               |
| 3    | Governance policy                            | None, moderate (50/50 capacity split with task-sensitivity routing), strict (70/30 split with trust-gated access, threshold $\geq 0.75$ ) | Graph: tree and entangled; load: medium and high; $N = 50$                                                                |
| 4    | Architecture $\times$ agents                 | Naïve vs. hybrid; $N \in \{20, 40, 60, 80\}$ ; per-agent arrival rate fixed (total demand scales with $N$ )                               | Graph: SP and entangled; load: medium and high                                                                            |
| 5    | Architecture $\times$ governance             | Naïve vs. hybrid (full); none vs. strict governance; tree, SP, entangled topologies                                                       | $N = 50$ ; load: medium and high; $2 \times 2 \times 3 \times 2 \times 10 = 240$ runs                                     |
| 6    | Allocation mechanism                         | Random, EDF, value-greedy (no prices), market (tâtonnement); naïve and hybrid architecture                                                | Tree, SP, entangled; load: medium and high; $N = 50$ ; no governance; $4 \times 3 \times 2 \times 2 \times 10 = 480$ runs |

TABLE V  
EXPERIMENT 1: IMPACT OF DAG TOPOLOGY AND LOAD LEVEL

| Topology  | Load   | Med. Lat. (ms) | $p_{95}$ Lat. (ms) | Drop Rate | Utilisation | Price Vol. ( $\sigma$ ) |
|-----------|--------|----------------|--------------------|-----------|-------------|-------------------------|
| Linear    | Low    | 126            | 145                | 0.34      | 0.14        | 0.000                   |
| Linear    | Medium | 130            | 150                | 0.35      | 0.29        | 0.000                   |
| Linear    | High   | 134            | 154                | 0.37      | 0.43        | 0.000                   |
| Tree      | Low    | 136            | 156                | 0.39      | 0.17        | 0.000                   |
| Tree      | Medium | 138            | 159                | 0.40      | 0.35        | 0.000                   |
| Tree      | High   | 142            | 163                | 0.49      | 0.52        | 0.000                   |
| SP        | Low    | 137            | 157                | 0.39      | 0.28        | 0.000                   |
| SP        | Medium | 140            | 160                | 0.53      | 0.55        | 0.000                   |
| SP        | High   | 146            | 168                | 0.90      | 0.83        | 0.112                   |
| Entangled | Low    | 143            | 164                | 0.44      | 0.37        | 0.000                   |
| Entangled | Medium | 149            | 170                | 0.77      | 0.74        | 0.373                   |
| Entangled | High   | 404            | 473                | 1.00      | 1.11        | 0.273                   |

Values averaged over 10 seeds. All topologies achieve comparable latency at low load (126–143 ms) with zero price volatility. Degradation under medium and high load is topology-dependent.

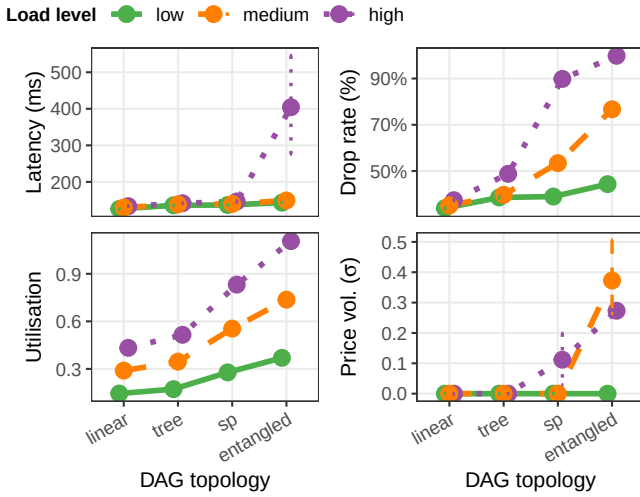


Fig. 1. Experiment 1: Effect of DAG topology and load on median latency (top-left), drop rate (top-right), utilisation (bottom-left), and price volatility (bottom-right). Polymatroidal topologies (linear, tree) maintain stable prices; entangled DAGs degrade sharply under load.

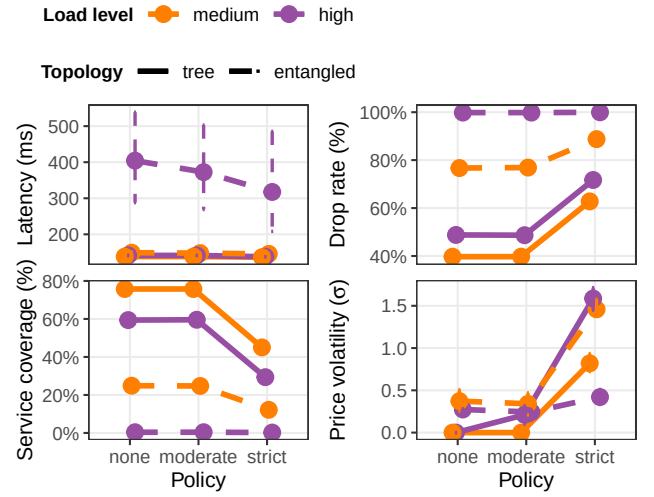


Fig. 2. Experiment 3: Effect of governance policy (none/moderate/strict) on performance for tree and entangled DAGs under medium and high load. Panels: median latency, drop rate, service coverage, price volatility.

TABLE VI  
EXPERIMENT 2: TOPOLOGY-DEPENDENT SCALING ( $N \in \{10, \dots, 60\}$ , MEDIUM LOAD)

| Topology  | $N$ | Med. Lat. (ms) | Drop Rate | Deadline Sat. | Price Vol. ( $\sigma$ ) |
|-----------|-----|----------------|-----------|---------------|-------------------------|
| Tree      | 10  | 136            | 0.38      | 0.62          | 0.000                   |
| Tree      | 20  | 136            | 0.38      | 0.62          | 0.000                   |
| Tree      | 30  | 137            | 0.39      | 0.61          | 0.000                   |
| Tree      | 40  | 138            | 0.40      | 0.60          | 0.000                   |
| Tree      | 50  | 138            | 0.40      | 0.60          | 0.000                   |
| Tree      | 60  | 139            | 0.40      | 0.60          | 0.000                   |
| SP        | 10  | 136            | 0.38      | 0.62          | 0.000                   |
| SP        | 20  | 138            | 0.40      | 0.60          | 0.000                   |
| SP        | 30  | 139            | 0.40      | 0.60          | 0.000                   |
| SP        | 40  | 140            | 0.42      | 0.58          | 0.000                   |
| SP        | 50  | 140            | 0.53      | 0.47          | 0.000                   |
| SP        | 60  | 139            | 0.67      | 0.33          | 0.000                   |
| Entangled | 10  | 142            | 0.43      | 0.57          | 0.000                   |
| Entangled | 20  | 145            | 0.46      | 0.54          | 0.000                   |
| Entangled | 30  | 148            | 0.49      | 0.51          | 0.116                   |
| Entangled | 40  | 149            | 0.63      | 0.37          | 0.287                   |
| Entangled | 50  | 149            | 0.77      | 0.23          | 0.373                   |
| Entangled | 60  | 152            | 0.93      | 0.07          | 0.301                   |

Values averaged over 10 seeds. Tree topology shows graceful scaling with near-constant metrics. Entangled topology reaches market failure by  $N = 60$ .

TABLE VII  
EXPERIMENT 3: GOVERNANCE AND TRUST EFFECTS (TREE AND ENTANGLED, MEDIUM AND HIGH LOAD)

| Topology  | Load   | Governance | Med. Lat. (ms) | Drop Rate | Coverage | Price Vol. ( $\sigma$ ) |
|-----------|--------|------------|----------------|-----------|----------|-------------------------|
| Tree      | Medium | None       | 138            | 0.40      | 0.76     | 0.000                   |
| Tree      | Medium | Moderate   | 138            | 0.40      | 0.76     | 0.000                   |
| Tree      | Medium | Strict     | 137            | 0.63      | 0.45     | 0.882                   |
| Tree      | High   | None       | 142            | 0.49      | 0.59     | 0.000                   |
| Tree      | High   | Moderate   | 142            | 0.49      | 0.60     | 0.246                   |
| Tree      | High   | Strict     | 138            | 0.72      | 0.30     | 1.686                   |
| Entangled | Medium | None       | 149            | 0.77      | 0.25     | 0.373                   |
| Entangled | Medium | Moderate   | 148            | 0.77      | 0.25     | 0.347                   |
| Entangled | Medium | Strict     | 146            | 0.89      | 0.12     | 1.338                   |
| Entangled | High   | None       | 404            | 1.00      | 0.004    | 0.273                   |
| Entangled | High   | Moderate   | 347            | 1.00      | 0.004    | 0.259                   |
| Entangled | High   | Strict     | 266            | 1.00      | 0.002    | 0.433                   |

Values averaged over 10 seeds. All allocated tasks are policy-compliant by construction. Strict governance substantially increases price volatility due to capacity partitioning into smaller sub-pools.

tree topologies, which exhibit zero volatility without governance but significant volatility under strict partitioning. This does not contradict polymatroidal preservation: governance-induced capacity partitioning pushes per-pool utilisation toward saturation, where the tâtonnement fails to converge despite equilibrium existence. The observation demonstrates that governance can introduce *operational* market instability even in structurally well-behaved topologies.

*Governance design-space considerations.*: The capacity-partitioning model tested above captures the structural effect of policy constraints on feasibility but represents only one point in a broader design space. In practice, governance may involve *data-locality restrictions* that constrain which tiers are admissible for specific task classes (e.g., medical data confined to jurisdiction-compliant nodes), *cross-domain policy negotiation* where integrators must reconcile conflicting jurisdictional rules before admitting cross-domain slices, and *dynamic trust*

*evolution* where reputation scores respond to adversarial behaviour or changing compliance requirements. The simulation results establish that even a relatively simple governance model produces quantifiable efficiency–compliance trade-offs that depend jointly on topology and load; richer models would likely amplify these interactions. This complements findings in trustworthy distributed systems [23], [24] by providing a resource- and dependency-aware perspective.

### C. Experiment 4: Architecture Ablation

Three strategies are compared: *naïve* (per-tier pricing), *hybrid (EMA only)* (smoothed slice price, efficiency = 1.0), and *hybrid (full)* (EMA plus efficiency factor: SP 0.75, entangled 0.85).

Figs. 3 and 4 present results in four panels. The price-volatility panel directly tests the encapsulation prediction.

TABLE VIII  
EXPERIMENT 4: NAÏVE VS. HYBRID ARCHITECTURE (SP AND ENTANGLED, MEDIUM AND HIGH LOAD)

| Topology  | Load   | Arch.  | $N$ | Med. Lat. (ms) | Drop Rate | Welfare | Price Vol. ( $\sigma$ ) |
|-----------|--------|--------|-----|----------------|-----------|---------|-------------------------|
| SP        | Medium | Naïve  | 20  | 138            | 0.40      | 9.2     | 0.000                   |
| SP        | Medium | Naïve  | 40  | 140            | 0.42      | 17.4    | 0.000                   |
| SP        | Medium | Naïve  | 60  | 139            | 0.67      | 14.8    | 0.000                   |
| SP        | Medium | Naïve  | 80  | 149            | 0.95      | 2.9     | 0.198                   |
| SP        | Medium | Hybrid | 20  | 137            | 0.39      | 9.3     | 0.096                   |
| SP        | Medium | Hybrid | 40  | 138            | 0.41      | 17.9    | 0.096                   |
| SP        | Medium | Hybrid | 60  | 138            | 0.67      | 15.1    | 0.096                   |
| SP        | Medium | Hybrid | 80  | 139            | 0.95      | 3.0     | 0.096                   |
| SP        | High   | Naïve  | 20  | 141            | 0.42      | 12.9    | 0.000                   |
| SP        | High   | Naïve  | 40  | 141            | 0.68      | 14.4    | 0.000                   |
| SP        | High   | Naïve  | 60  | 197            | 0.99      | 0.3     | 0.338                   |
| SP        | High   | Naïve  | 80  | 635            | 1.00      | 0.0     | 0.286                   |
| SP        | High   | Hybrid | 20  | 139            | 0.41      | 13.4    | 0.096                   |
| SP        | High   | Hybrid | 40  | 139            | 0.67      | 14.9    | 0.096                   |
| SP        | High   | Hybrid | 60  | 181            | 0.99      | 0.4     | 0.096                   |
| SP        | High   | Hybrid | 80  | 674            | 1.00      | 0.0     | 0.097                   |
| Entangled | Medium | Naïve  | 20  | 145            | 0.46      | 8.0     | 0.000                   |
| Entangled | Medium | Naïve  | 40  | 149            | 0.63      | 10.8    | 0.287                   |
| Entangled | Medium | Naïve  | 60  | 152            | 0.93      | 3.1     | 0.301                   |
| Entangled | Medium | Naïve  | 80  | 439            | 1.00      | 0.0     | 0.274                   |
| Entangled | Medium | Hybrid | 20  | 144            | 0.45      | 8.2     | 0.096                   |
| Entangled | Medium | Hybrid | 40  | 146            | 0.61      | 11.4    | 0.096                   |
| Entangled | Medium | Hybrid | 60  | 155            | 0.93      | 2.9     | 0.154                   |
| Entangled | Medium | Hybrid | 80  | 473            | 1.00      | 0.0     | 0.097                   |
| Entangled | High   | Naïve  | 20  | 151            | 0.53      | 10.1    | 0.066                   |
| Entangled | High   | Naïve  | 40  | 157            | 0.93      | 3.0     | 0.307                   |
| Entangled | High   | Naïve  | 60  | 683            | 1.00      | 0.0     | 0.270                   |
| Entangled | High   | Naïve  | 80  | 688            | 1.00      | 0.0     | 0.346                   |
| Entangled | High   | Hybrid | 20  | 147            | 0.50      | 10.9    | 0.096                   |
| Entangled | High   | Hybrid | 40  | 162            | 0.93      | 3.0     | 0.097                   |
| Entangled | High   | Hybrid | 60  | 684            | 1.00      | 0.0     | 0.097                   |
| Entangled | High   | Hybrid | 80  | 684            | 1.00      | 0.0     | 0.097                   |

Values averaged over 10 seeds. Hybrid architecture reduces price volatility by up to 70–75%. Both architectures saturate similarly beyond capacity ( $N \geq 60$  for entangled/high,  $N \geq 80$  for SP/high).

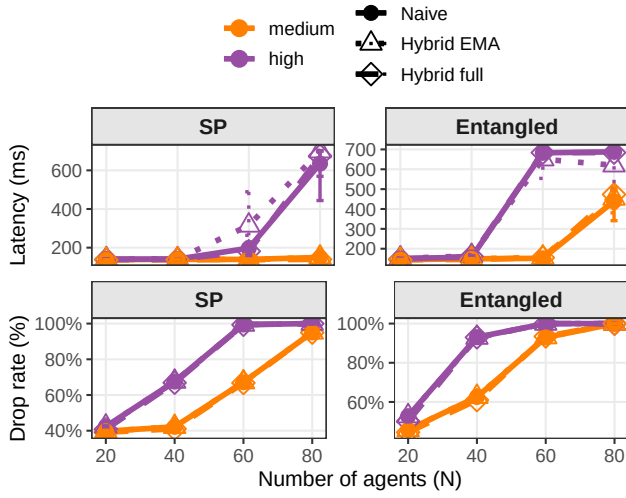


Fig. 3. Experiment 4 (operational): Naïve, hybrid (EMA only), and hybrid (full) architecture across agent counts, DAG types, and loads. Panels: latency, drop rate.

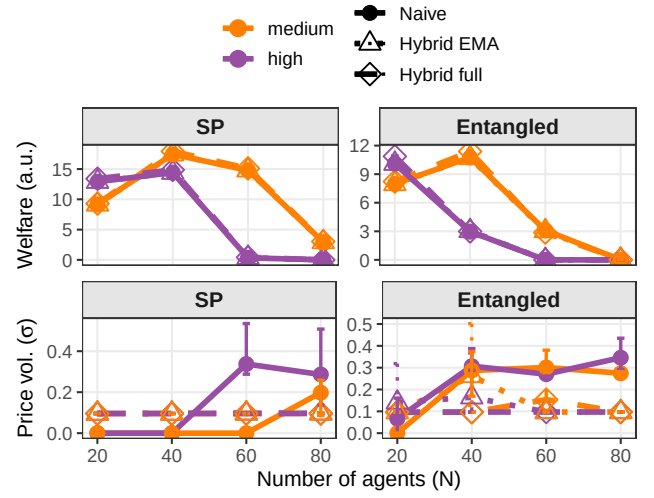


Fig. 4. Experiment 4 (economic): Welfare and price volatility under the same conditions as Fig. 3.

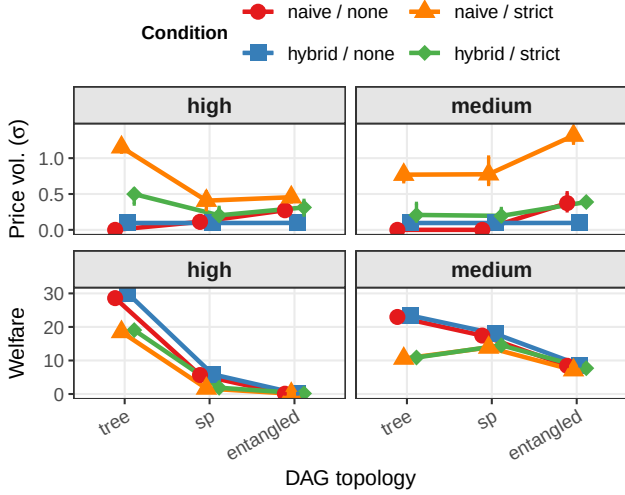


Fig. 5. Experiment 5: Architecture  $\times$  governance interaction. Top row: price volatility by topology and condition; bottom row: welfare. The hybrid architecture mitigates the volatility penalty introduced by governance.

Under naïve allocation, price volatility grows with  $N$  and load ( $\sigma = 0.35$  for entangled at  $N = 80$ /high,  $\sigma = 0.34$  for SP at  $N = 60$ /high). Under hybrid allocation, volatility remains at or below  $\sigma = 0.10$  in nearly all conditions, a reduction of 70–75%.

For latency, hybrid provides modest but consistent improvements (SP/high at  $N = 60$ : 181 ms vs. 197 ms for naïve). Both architectures saturate at similar thresholds ( $N \approx 60$  for entangled/high,  $N \approx 80$  for SP/high), reaching median latencies of 635–688 ms with near-total drop rates, confirming that the hybrid architecture’s primary contribution is price stability rather than raw latency reduction.

An EMA-only ablation confirms that EMA smoothing accounts for the majority of volatility reduction (SP/high/ $N = 60$ :  $\sigma = 0.097$  vs. naïve 0.337), while latency and welfare improvements depend on the efficiency factor (315 ms EMA-only vs. 181 ms full hybrid). The two serve distinct roles: EMA stabilises prices (architectural), while the efficiency factor reduces congestion (operational).

#### D. Experiment 5: Architecture $\times$ Governance Interaction

This experiment crosses architecture (naïve / hybrid)  $\times$  governance (none / strict)  $\times$  topology (tree / SP / entangled)  $\times$  load (medium / high), yielding 240 runs.

Fig. 5 presents price volatility and welfare across the four condition combinations. When both the hybrid architecture and strict governance are active, the hybrid architecture substantially mitigates the governance-induced volatility penalty: for SP under medium load,  $\sigma$  drops from 0.78 (naïve/strict) to 0.20 (hybrid/strict), a 75% reduction. For tree/high, hybrid/strict reduces volatility from  $\sigma = 1.15$  to  $\sigma = 0.50$  (57%). For entangled/medium,  $\sigma = 1.32$  to  $\sigma = 0.39$  (70%).

Welfare results reveal the interaction’s nature. For SP topologies, the combination is approximately additive (synergy CI includes zero). For tree/high, the interaction is significantly

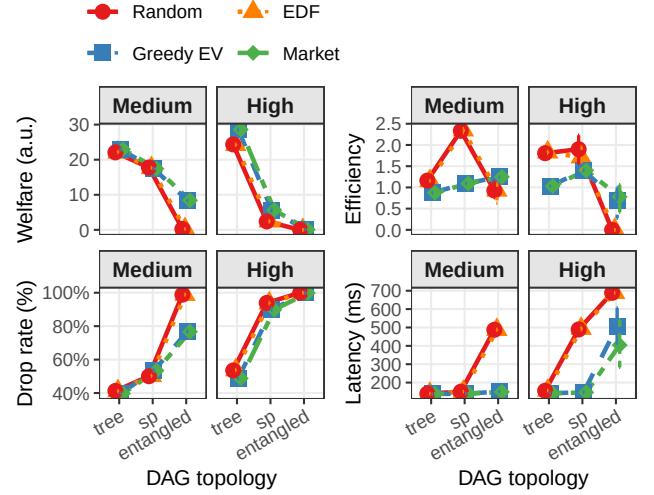


Fig. 6. Experiment 6: Mechanism ablation comparing random, EDF, value-greedy, and market allocation across topologies and load levels (naïve architecture). Panels: welfare (top-left), efficiency ratio (top-right), drop rate (bottom-left), median latency (bottom-right).

sub-additive ( $\Delta_{\text{synergy}} = -0.83 [-1.04, -0.59]$ ): governance reduces welfare more sharply when hybrid is active because the efficiency factor leaves less slack. For entangled topologies, the interaction is mildly super-additive (0.39 [0.18, 0.64]).

#### E. Experiment 6: Mechanism Ablation

Four allocation rules are compared: random feasible packing, earliest-deadline-first (EDF), value-greedy (packing by  $\mathbb{E}[V]$  without prices), and the full tâtonnement market mechanism. Each level adds one component (scheduling awareness, value model, price signals).

Fig. 6 presents results for the naïve architecture. The market and value-greedy curves overlap almost exactly across all conditions (welfare difference  $< 1\%$ ), because tâtonnement prices converge to near-zero under truthful bidding.

The mechanism ordering is topology-dependent. For tree/high load, value-greedy/market achieve welfare of 28.6 a.u. vs. 24.3–24.4 for random/EDF (17% improvement). For entangled/medium, the value model yields 8.4 a.u. vs. 0.26 for random/EDF (32 $\times$ ), as it identifies tasks with sufficient deadline slack to survive high-congestion paths. For SP/medium, all mechanisms perform comparably (17.4–17.8 a.u.), with random and EDF marginally outperforming.

Under hybrid architecture, the SP reversal is amplified at high load (random/EDF: 18.2–18.3 a.u. vs. value-greedy/market: 5.9 a.u.), because the efficiency factor creates headroom where load-spreading heuristics outperform value concentration.

## VI. STATISTICAL ANALYSIS

This section describes the statistical methodology used throughout the evaluation and presents detailed test results for each experiment. All analyses use non-parametric methods

appropriate for small-sample simulation studies ( $n = 10$  seeds per condition).

#### A. Methodology

*Confidence intervals.*: All reported metrics use bias-corrected and accelerated (BCa) bootstrap 95% confidence intervals with  $B = 2,000$  resamples over the 10 seeds per condition [29]. For conditions where the BCa method fails to converge (e.g., zero-variance metrics), we fall back to the percentile method.

*Hypothesis tests.*: For multi-group comparisons (e.g., topology with 4 levels), we use the Kruskal–Wallis  $H$ -test [30], a non-parametric analogue of one-way ANOVA. Post-hoc pairwise comparisons use two-sided Wilcoxon rank-sum (Mann–Whitney  $U$ ) tests with Holm correction for multiple comparisons [31]. The choice of non-parametric tests is motivated by the small sample size ( $n = 10$ ), which precludes reliable normality assessment, and by the nature of simulation metrics (latency, drop rate, price volatility), which are typically non-normal with skewed or bounded distributions.

*Effect sizes.*: Pairwise effect sizes are reported as Cliff’s delta ( $\delta$ ), a non-parametric measure appropriate for ordinal and non-normal data [32]. Magnitude thresholds follow Romano et al. [33]:  $|\delta| < 0.147$  (negligible),  $< 0.33$  (small),  $< 0.474$  (medium),  $\geq 0.474$  (large).

*Interaction analysis.*: Factorial interactions (e.g., topology  $\times$  load in Exp 1, architecture  $\times$  governance in Exp 5) are tested using Aligned Rank Transform (ART) ANOVA [34], a non-parametric method that supports multi-factor interaction analysis without normality assumptions. We report  $F$ -statistics and  $p$ -values for main effects and two-way interactions.

*Synergy analysis.*: For Experiment 5, we assess whether the combination of hybrid architecture and governance yields super-additive (synergistic) or sub-additive (redundant) effects. Let  $W_{ij}$  denote the mean welfare under architecture  $i \in \{\text{naïve}, \text{hybrid}\}$  and governance  $j \in \{\text{none}, \text{strict}\}$ . The synergy measure is:

$$\Delta_{\text{synergy}} = \underbrace{(W_{\text{hybrid}, \text{strict}} - W_{\text{naïve}, \text{none}})}_{\text{joint gain}} - \underbrace{(W_{\text{hybrid}, \text{none}} - W_{\text{naïve}, \text{none}})}_{\text{marginal arch.}} - \underbrace{(W_{\text{naïve}, \text{strict}} - W_{\text{naïve}, \text{none}})}_{\text{marginal gov.}} \quad (1)$$

Bootstrap 95% CIs are computed for  $\Delta_{\text{synergy}}$ ; if the interval includes zero, the composition is additive.

#### B. Experiment 1: Statistical Summary

Kruskal–Wallis tests confirm that topology has a highly significant main effect on all key metrics at every load level. At high load, the effect of topology on median latency ( $H(3) = 36.6$ ,  $p < 10^{-7}$ ), drop rate ( $H(3) = 36.6$ ,  $p < 10^{-7}$ ), welfare ( $H(3) = 36.6$ ,  $p < 10^{-7}$ ), and price volatility ( $H(3) = 26.3$ ,  $p < 10^{-5}$ ) is significant. At medium load, all four metrics remain significant ( $p < 10^{-7}$  for latency, drop rate, and welfare;  $p < 10^{-7}$  for volatility). At low load, topology still significantly affects latency, drop rate, and welfare ( $p < 10^{-7}$ ),

though price volatility is identically zero across all topologies (no test applicable).

Pairwise Wilcoxon tests (Holm-corrected) show that the entangled topology differs significantly from all other topologies on all metrics at medium and high load ( $p_{\text{adj}} < 0.002$ ). The ordering linear  $\approx$  tree  $<$  SP  $<$  entangled is confirmed by the pairwise test structure: all entangled-vs-other comparisons are significant, while linear-vs-tree comparisons are significant but with smaller effect magnitudes.

#### C. Experiment 2: Statistical Summary

Spearman rank correlations between agent count ( $N$ ) and each metric confirm topology-dependent scaling. For the tree topology, all metrics show strong monotonic relationships with  $N$  ( $\rho = 0.78\text{--}0.99$ ,  $p < 10^{-13}$ ), but the absolute magnitudes of change are small (e.g., median latency increases by only 3 ms across  $N = 10\text{--}60$ ). For SP, correlations are similarly strong ( $\rho = 0.73\text{--}0.99$ ,  $p < 10^{-10}$ ), with price volatility remaining at zero throughout (correlation not applicable). For the entangled topology, the correlations are also strong ( $\rho > 0.7$ ,  $p < 10^{-10}$  for latency, drop rate, utilisation, and welfare), confirming the steep monotonic degradation visible in the figures. Price volatility for the entangled topology shows a non-monotonic pattern (rising then falling), reflecting the transition from stable to fully saturated operation where price oscillations collapse.

#### D. Experiment 3: Statistical Summary

Kruskal–Wallis tests show that governance policy has a significant effect on all metrics for the tree topology under high load: median latency ( $H(2) = 19.4$ ,  $p < 10^{-4}$ ), drop rate ( $H(2) = 19.4$ ,  $p < 10^{-4}$ ), welfare ( $H(2) = 19.4$ ,  $p < 10^{-4}$ ), coverage ( $H(2) = 19.4$ ,  $p < 10^{-4}$ ), and price volatility ( $H(2) = 24.1$ ,  $p < 10^{-5}$ ). Pairwise Wilcoxon tests (Holm-corrected) confirm that the none-vs-strict contrast is significant for all metrics ( $p_{\text{adj}} < 0.001$ ).

For the entangled topology under high load, governance effects are more nuanced. Welfare ( $H(2) = 6.10$ ,  $p = 0.048$ ) and coverage ( $H(2) = 8.72$ ,  $p = 0.013$ ) show significant effects, but median latency does not ( $H(2) = 1.86$ ,  $p = 0.39$ ), reflecting the fact that the system is already saturated regardless of governance policy. Price volatility shows a significant governance effect ( $H(2) = 11.0$ ,  $p = 0.004$ ), confirming that capacity partitioning amplifies price oscillations even in structurally unstable topologies.

#### E. Experiment 4: Statistical Summary

Kruskal–Wallis tests for the effect of architecture (naïve, EMA-only, full hybrid) show that the primary significant effect is on price volatility, not on latency or welfare directly. For entangled/medium load, architecture significantly affects price volatility ( $H(2) = 11.1$ ,  $p = 0.004$ ) but not latency ( $H(2) = 2.74$ ,  $p = 0.25$ ) or welfare ( $H(2) = 0.72$ ,  $p = 0.70$ ). For SP/high load, architecture has a marginally significant effect on latency ( $H(2) = 6.14$ ,  $p = 0.046$ ) but not on drop rate ( $p = 0.69$ ) or welfare ( $p = 0.57$ ). This confirms that the hybrid architecture’s primary contribution is price stabilisation

rather than raw throughput improvement, consistent with the main text’s interpretation that the benefit concentrates in the congested-but-not-saturated regime.

#### F. Experiment 5: Statistical Summary

**Main effects.**: Kruskal–Wallis tests across all 240 runs confirm that governance policy has a highly significant main effect on all metrics: latency ( $H(1) = 16.8$ ,  $p < 10^{-4}$ ), drop rate ( $H(1) = 15.6$ ,  $p < 10^{-4}$ ), welfare ( $H(1) = 10.5$ ,  $p = 0.001$ ), and price volatility ( $H(1) = 118$ ,  $p < 10^{-26}$ ). Architecture has a significant main effect on latency ( $H(1) = 10.1$ ,  $p = 0.001$ ) and price volatility ( $H(1) = 10.3$ ,  $p = 0.001$ ), but not on drop rate ( $p = 0.57$ ) or welfare ( $p = 0.52$ ), confirming that the hybrid architecture acts primarily through price stabilisation.

**Synergy analysis.**: The synergy measure  $\Delta_{\text{synergy}}$  (see Section VI methodology) reveals topology-dependent interaction patterns. For the tree topology under high load,  $\Delta_{\text{synergy}} = -0.83$  [95% CI:  $-1.04, -0.59$ ], indicating significant sub-additivity: the combination achieves less than the sum of individual gains. For tree/medium,  $\Delta_{\text{synergy}} = -0.13$   $[-0.38, 0.10]$ , with the CI including zero (additive). For SP topologies, the synergy term is near zero and CIs include zero in both load conditions (SP/medium:  $0.10$   $[-0.21, 0.48]$ ; SP/high:  $0.01$   $[-0.31, 0.24]$ ), confirming additive composition. For entangled topologies, the synergy term is positive with CIs excluding zero (entangled/medium:  $0.39$   $[0.18, 0.64]$ ; entangled/high:  $0.08$   $[0.004, 0.24]$ ), suggesting mildly super-additive effects where the combination of price smoothing and governance partitioning provides more benefit than predicted by individual contributions, though the absolute magnitude is small given that entangled/high is near total saturation.

#### G. Experiment 6: Mechanism Ablation Results

Experiment 6 compares four allocation mechanisms (random, EDF, value-greedy, market) across three topologies, two load levels, and two architectures ( $4 \times 3 \times 2 \times 2 \times 10 = 480$  runs). The main text presents naïve architecture results; here we report the full result table and statistical summary including hybrid architecture conditions.

**Key finding: market  $\approx$  value-greedy.**: Across all 480 runs, the market mechanism and value-greedy allocation produce near-identical outcomes: the maximum welfare difference is 0.07 a.u. ( $< 1\%$  of peak welfare), and the maximum drop-rate difference is 0.16 percentage points. This occurs because tâtonnement prices converge to near-zero under truthful bidding: with no strategic misreporting, the price signal is informationally redundant and surplus ordering reduces to value ordering.

**Hybrid architecture.**: The market–value-greedy near-equivalence holds under hybrid architecture. A notable architecture-dependent result appears for SP/high load: under hybrid, random and EDF allocation achieve welfare of 18.2–18.3 a.u. versus 5.9 for value-greedy/market. The hybrid architecture’s efficiency factor reduces effective demand, creating headroom where load-spreading heuristics outperform value-concentrating strategies. This reversal does not occur under

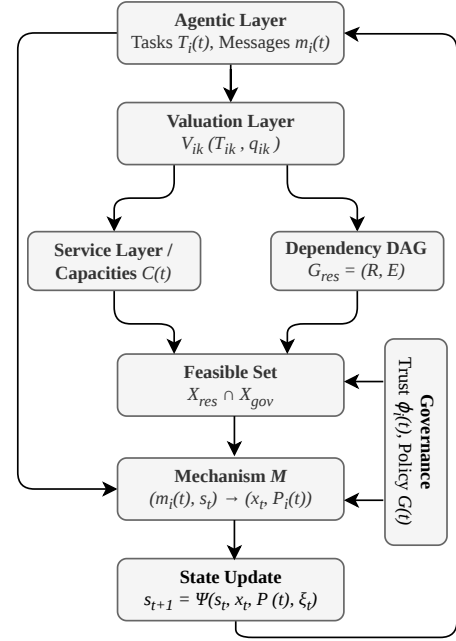


Fig. 7. Model overview: agentic layer, latency-aware valuations, resource dependencies, governance constraints, mechanism, and state evolution.

naïve architecture at the same load, demonstrating that the optimal allocation strategy depends jointly on topology, load, and architectural context.

#### H. Experiment 6: Statistical Summary

**Main effects (naïve architecture).**: Kruskal–Wallis tests for the mechanism factor are highly significant in all topology  $\times$  load conditions when stratified by architecture (all  $p < 0.002$ ). The strongest effects appear for entangled/medium ( $H(3) = 29.4$ ,  $p < 10^{-6}$ ) and tree/high ( $H(3) = 29.5$ ,  $p < 10^{-6}$ ).

**Pairwise comparisons (Wilcoxon, Holm-corrected).**: The greedy\_ev vs. market contrast is non-significant in every condition ( $p_{\text{adj}} \geq 0.59$ ; Cliff’s  $\delta \leq 0.15$ , negligible), confirming they are statistically indistinguishable. The greedy\_ev vs. random contrast is significant and large in most conditions: tree/high ( $p = 0.001$ ,  $\delta = 1.0$ ), entangled/medium ( $p = 0.001$ ,  $\delta = 1.0$ ), SP/high ( $p = 0.001$ ,  $\delta = 1.0$ ). The SP/medium reversal (random outperforms greedy\_ev) is also statistically significant ( $p = 0.022$ ,  $\delta = -0.78$ , large effect). EDF and random are not distinguishable in any condition ( $p \geq 0.24$ ).

**Theory-to-experiment mapping.**: Table XII links each formal result from the main paper to the experiment that tests its predictions and the key observables used for evaluation.

**Ablation table (complete).**: The following table summarises the full ablation matrix across all six experiments. Each row indicates which components are ablated and the corresponding experiment.

## VII. ADDITIONAL MATERIAL

### A. Model Overview and Assumptions

The framework in the main paper rests on assumptions standard in network/service management and mechanism design.

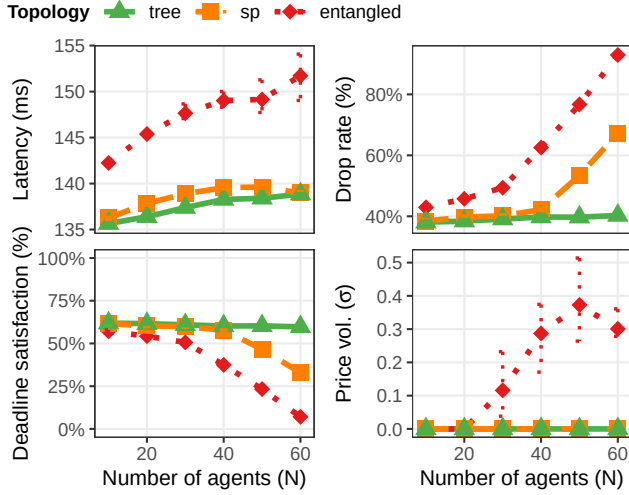


Fig. 8. Experiment 2: Topology-dependent scaling as agent count increases ( $N = 10$ – $60$ ). Panels: median latency, drop rate, deadline satisfaction, price volatility.

In particular, we assume that: (i) autonomous agents drive service demand and resource negotiation [35], [19]; (ii) tasks are latency- and QoS-sensitive [36]; (iii) the system spans heterogeneous device–edge–cloud resources [37], [36]; (iv) services depend on multi-layer resource compositions [36], [16]; (v) resource availability and network conditions vary over time [36], [38]; (vi) governance and policies constrain resource usage [22]; (vii) agents may withhold or misreport private information [25]; (viii) reputation reflects provider reliability [23], [24]; (ix) architectural encapsulation (slicing, virtualisation) is possible [39], [16]; and (x) at each decision epoch, agents evaluate allocations with respect to a sufficiently aligned representation of the system state [40], [41]. That is, while private information and strategic incentives remain, agents condition their valuations on a commonly accepted state description  $s_t$  capturing resource capacities, network conditions, governance constraints, and reputational signals. The paper therefore focuses on normative allocation and mechanism design given a shared operational state, and does not model heterogeneous beliefs or epistemic disagreement about the underlying environment. The system state evolves in a Markovian manner driven by allocations, network conditions, and stochastic events.

### B. Experiment 2: Topology-Dependent Scaling

The second experiment tests whether the rate of performance degradation as the agent population increases depends on DAG topology, reinforcing that structure is a first-order determinant of scalability. We compare three topologies (tree, SP, entangled) across  $N \in \{10, 20, 30, 40, 50, 60\}$  agents under medium load ( $\lambda = 1.0$ ), with naïve architecture and no governance. This yields  $3 \times 6 \times 10 = 180$  runs.

Fig. 8 reveals three qualitatively different scaling regimes. The tree topology exhibits graceful degradation: median latency increases by only 3 ms across the full range (136 ms at  $N = 10$  to 139 ms at  $N = 60$ ), drop rates rise gently from

38% to 40%, deadline satisfaction remains approximately 60% throughout, and price volatility stays at zero. The balanced per-tier demand of tree-structured tasks distributes load evenly across tiers, preventing any single tier from saturating.

The SP topology shows moderate degradation. Latency and drop rates remain stable up to approximately  $N = 40$  (median latency 140 ms, drop rate 42%), then increase as cloud-tier demand exceeds capacity: by  $N = 60$ , the drop rate reaches 67% and deadline satisfaction falls to 33%. However, prices remain stable (zero volatility) even at  $N = 60$ , indicating that the SP topology’s fully polymatroidal structure (Proposition 1) supports convergent tâtonnement when utilisation remains below the saturation threshold.

The entangled topology degrades earliest and most sharply. Drop rates rise steeply from 43% at  $N = 10$  to 93% at  $N = 60$ , and deadline satisfaction falls from 57% to just 7%. Critically, price volatility emerges at  $N = 30$  ( $\sigma = 0.116$ ) and peaks at  $\sigma = 0.373$  around  $N = 50$ , confirming that the onset of price instability coincides with the point at which per-tier demand exceeds the capacity threshold. Median latency for the entangled topology increases from 142 ms to 152 ms across the range, a modest absolute increase that obscures the far more dramatic degradation visible in drop rate and price stability.

These results show that the “scaling ceiling”, the agent count at which the system transitions from stable to degraded operation, depends critically on topology. For the tree topology, no such ceiling is visible within the tested range. For SP, degradation begins around  $N = 40$ . For entangled, degradation begins by  $N = 20$ – $30$ , with complete market failure (near-total drops, persistent price oscillation) by  $N = 60$ . This confirms that topology is a first-order determinant of scalability, not merely a secondary factor behind load.

### C. Extended Mechanism-Design Discussion

This section provides additional detail on three aspects of the mechanism-design results that are condensed in the main paper for space.

#### *Governance preservation of polymatroidal structure.*

Governance constraints  $\mathcal{X}_{\text{gov}}$  impose coordinate-wise upper bounds  $x_l \leq u_l$  on each leaf service, encoding trust-based admission, jurisdictional filters, and access control. The intersection of a polymatroid  $P(f)$  with coordinate truncations  $\{x : x_l \leq u_l\}$  is again a polymatroid, with rank function  $f'(S) = \min_{T \subseteq S} \{f(T) + \sum_{l \in S \setminus T} u_l\}$  [2]. Hence  $\mathcal{X}_t = \mathcal{X}_{\text{res}} \cap \mathcal{X}_{\text{gov}}$  inherits polymatroidal structure. This preservation result relies on governance constraints being expressible as independent per-leaf upper bounds; more complex governance couplings (e.g., mutual exclusivity across leaves, cross-jurisdictional budget quotas, or path-specific exclusions) would generally not preserve polymatroidal structure and require separate treatment.

*Conditions for gross-substitutes valuations.* Condition (c) of Lemma 1 ensures that the items over which GS is defined have stable characteristics during each allocation round; between epochs, slice attributes may change as congestion and routing conditions evolve, and GS is re-established under the updated parameters. Note that slice capacities (the

number of available units of each slice type) are determined by the polymatroid  $\mathcal{X}_{\text{res}}$ , not by the valuations; this is a feasibility condition relevant to Proposition 2 rather than a requirement for GS itself. Additive separability (condition (b)) is critical: shared agent-side budgets, cross-task latency coupling, or shared token caps would break it and can destroy the GS property. Without encapsulation, agents would bid on resource bundles along DAG paths, introducing complementarities that violate GS. Encapsulation eliminates these by absorbing them into the integrator.

*Roles of polymatroidal structure and inter-epoch limitations.*: Together, Propositions 1–2 and Lemma 1 establish that, *within each decision epoch* (during which slice attributes are exogenous and fixed), real-time AI service economies with tree or series–parallel dependency structures admit mechanisms that are efficient, DSIC, and individually rational. The polymatroidal structure serves three distinct roles in this chain: characterising the feasibility geometry of the supply region (Proposition 1), enabling Walrasian equilibrium existence under GS valuations via discrete convex analysis (Proposition 2(i)), and supporting the clinching-auction implementation in the single-parameter case (Proposition 2(iii)). When agents’ demands decompose into independent per-task single-parameter demands, the ascending clinching auction additionally guarantees weak budget balance, with polynomial-time complexity that is practical when the number of slice types and the valuation range are moderate. Across epochs, load-induced changes to slice latencies and capacities may alter the items being traded; the mechanism is re-invoked at each epoch under updated parameters, and the GS/equilibrium guarantees apply epoch-by-epoch rather than as a dynamic-equilibrium result. In particular, the repeated interaction across epochs can induce strategic dynamics—such as learning, reputation gaming, or inter-epoch demand manipulation—that are outside the scope of the present per-epoch theorems.

*Theoretical positioning.*: The formal results in this paper apply classical polymatroid theory, the gross-substitutes characterisation, and VCG mechanism design to a novel application domain: real-time AI service economies across the computing continuum. The main contribution is problem formulation, not new mechanism-design theory: identifying that AI service-dependency graphs with tree or series–parallel structure satisfy the structural conditions (polymatroidal feasibility, gross substitutes) under which efficient, incentive-compatible mechanisms exist and are polynomial-time computable. Individual results (Propositions 1–2 and Lemma 1) instantiate known theorems in the service-dependency setting; the encapsulation result (Proposition 3) provides the paper’s main conceptual contribution by reducing cross-domain allocation to matroid feasibility, bridging classical single-domain mechanism design and the multi-tier architecture that production deployments require. The theoretical value lies in the structural identification (that the problem has the right shape for these classical results) and in the architectural consequence (that encapsulation makes the classical results operationally deployable), rather than in the individual mechanism-design results themselves.

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TABLE IX  
EXPERIMENT 4 ABLATION: NAÏVE, HYBRID (EMA ONLY), AND HYBRID (FULL) ARCHITECTURE COMPARISON

| Topology  | Load   | Architecture | $N$ | Med. Lat. (ms) | Drop Rate | Welfare | Price Vol. ( $\sigma$ ) |
|-----------|--------|--------------|-----|----------------|-----------|---------|-------------------------|
| SP        | Medium | Naïve        | 20  | 138            | 0.40      | 9.2     | 0.000                   |
| SP        | Medium | EMA only     | 20  | 138            | 0.40      | 9.2     | 0.096                   |
| SP        | Medium | Full hybrid  | 20  | 137            | 0.39      | 9.3     | 0.096                   |
| SP        | Medium | Naïve        | 40  | 140            | 0.42      | 17.4    | 0.000                   |
| SP        | Medium | EMA only     | 40  | 140            | 0.42      | 17.4    | 0.096                   |
| SP        | Medium | Full hybrid  | 40  | 138            | 0.41      | 17.9    | 0.096                   |
| SP        | Medium | Naïve        | 60  | 139            | 0.67      | 14.8    | 0.000                   |
| SP        | Medium | EMA only     | 60  | 139            | 0.67      | 14.8    | 0.096                   |
| SP        | Medium | Full hybrid  | 60  | 138            | 0.67      | 15.1    | 0.096                   |
| SP        | Medium | Naïve        | 80  | 149            | 0.95      | 2.9     | 0.198                   |
| SP        | Medium | EMA only     | 80  | 149            | 0.95      | 3.0     | 0.097                   |
| SP        | Medium | Full hybrid  | 80  | 139            | 0.95      | 3.0     | 0.096                   |
| SP        | High   | Naïve        | 20  | 141            | 0.42      | 12.9    | 0.000                   |
| SP        | High   | EMA only     | 20  | 141            | 0.42      | 12.9    | 0.096                   |
| SP        | High   | Full hybrid  | 20  | 139            | 0.41      | 13.4    | 0.096                   |
| SP        | High   | Naïve        | 40  | 141            | 0.68      | 14.4    | 0.000                   |
| SP        | High   | EMA only     | 40  | 141            | 0.68      | 14.4    | 0.096                   |
| SP        | High   | Full hybrid  | 40  | 139            | 0.67      | 14.9    | 0.096                   |
| SP        | High   | Naïve        | 60  | 197            | 1.00      | 0.3     | 0.337                   |
| SP        | High   | EMA only     | 60  | 315            | 0.99      | 0.4     | 0.097                   |
| SP        | High   | Full hybrid  | 60  | 181            | 0.99      | 0.4     | 0.096                   |
| SP        | High   | Naïve        | 80  | 635            | 1.00      | 0.0     | 0.286                   |
| SP        | High   | EMA only     | 80  | 699            | 1.00      | 0.0     | 0.097                   |
| SP        | High   | Full hybrid  | 80  | 674            | 1.00      | 0.0     | 0.097                   |
| Entangled | Medium | Naïve        | 20  | 145            | 0.46      | 8.0     | 0.000                   |
| Entangled | Medium | EMA only     | 20  | 145            | 0.46      | 8.0     | 0.096                   |
| Entangled | Medium | Full hybrid  | 20  | 144            | 0.45      | 8.2     | 0.096                   |
| Entangled | Medium | Naïve        | 40  | 149            | 0.63      | 10.8    | 0.287                   |
| Entangled | Medium | EMA only     | 40  | 150            | 0.63      | 10.7    | 0.262                   |
| Entangled | Medium | Full hybrid  | 40  | 146            | 0.61      | 11.4    | 0.096                   |
| Entangled | Medium | Naïve        | 60  | 152            | 0.93      | 3.1     | 0.301                   |
| Entangled | Medium | EMA only     | 60  | 156            | 0.93      | 3.0     | 0.097                   |
| Entangled | Medium | Full hybrid  | 60  | 155            | 0.93      | 2.9     | 0.154                   |
| Entangled | Medium | Naïve        | 80  | 439            | 1.00      | 0.0     | 0.274                   |
| Entangled | Medium | EMA only     | 80  | 453            | 1.00      | 0.0     | 0.097                   |
| Entangled | Medium | Full hybrid  | 80  | 473            | 1.00      | 0.0     | 0.097                   |
| Entangled | High   | Naïve        | 20  | 151            | 0.53      | 10.1    | 0.066                   |
| Entangled | High   | EMA only     | 20  | 151            | 0.53      | 10.1    | 0.141                   |
| Entangled | High   | Full hybrid  | 20  | 147            | 0.50      | 10.9    | 0.096                   |
| Entangled | High   | Naïve        | 40  | 157            | 0.93      | 3.0     | 0.307                   |
| Entangled | High   | EMA only     | 40  | 164            | 0.93      | 3.0     | 0.165                   |
| Entangled | High   | Full hybrid  | 40  | 162            | 0.93      | 3.0     | 0.097                   |
| Entangled | High   | Naïve        | 60  | 683            | 1.00      | 0.0     | 0.270                   |
| Entangled | High   | EMA only     | 60  | 651            | 1.00      | 0.0     | 0.097                   |
| Entangled | High   | Full hybrid  | 60  | 684            | 1.00      | 0.0     | 0.097                   |
| Entangled | High   | Naïve        | 80  | 688            | 1.00      | 0.0     | 0.346                   |
| Entangled | High   | EMA only     | 80  | 620            | 1.00      | 0.0     | 0.097                   |
| Entangled | High   | Full hybrid  | 80  | 684            | 1.00      | 0.0     | 0.097                   |

Values averaged over 10 seeds. EMA only: integrator with price smoothing but efficiency = 1.0; full hybrid: EMA plus efficiency factor (SP: 0.75, entangled: 0.85). EMA smoothing accounts for the majority of the price-volatility reduction; the efficiency factor primarily improves latency and welfare in the congested-but-not-saturated regime.

TABLE X  
EXPERIMENT 5: ARCHITECTURE  $\times$  GOVERNANCE INTERACTION ( $N = 50$ , MEDIUM AND HIGH LOAD)

| Topology  | Load   | Arch.  | Gov.   | Med. Lat. (ms) | Drop Rate | Welfare | Price Vol. ( $\sigma$ ) |
|-----------|--------|--------|--------|----------------|-----------|---------|-------------------------|
| Tree      | Medium | Naïve  | None   | 138            | 0.40      | 23.0    | 0.000                   |
| Tree      | Medium | Naïve  | Strict | 137            | 0.73      | 10.6    | 0.770                   |
| Tree      | Medium | Hybrid | None   | 137            | 0.39      | 23.4    | 0.096                   |
| Tree      | Medium | Hybrid | Strict | 136            | 0.72      | 10.9    | 0.208                   |
| Tree      | High   | Naïve  | None   | 142            | 0.49      | 28.6    | 0.000                   |
| Tree      | High   | Naïve  | Strict | 138            | 0.67      | 18.5    | 1.152                   |
| Tree      | High   | Hybrid | None   | 139            | 0.47      | 30.0    | 0.096                   |
| Tree      | High   | Hybrid | Strict | 137            | 0.67      | 19.1    | 0.498                   |
| SP        | Medium | Naïve  | None   | 140            | 0.53      | 17.4    | 0.000                   |
| SP        | Medium | Naïve  | Strict | 138            | 0.63      | 13.9    | 0.775                   |
| SP        | Medium | Hybrid | None   | 138            | 0.52      | 18.0    | 0.096                   |
| SP        | Medium | Hybrid | Strict | 137            | 0.62      | 14.5    | 0.195                   |
| SP        | High   | Naïve  | None   | 146            | 0.90      | 5.7     | 0.112                   |
| SP        | High   | Naïve  | Strict | 140            | 0.97      | 1.7     | 0.407                   |
| SP        | High   | Hybrid | None   | 140            | 0.90      | 5.9     | 0.096                   |
| SP        | High   | Hybrid | Strict | 139            | 0.97      | 1.9     | 0.202                   |
| Entangled | Medium | Naïve  | None   | 149            | 0.77      | 8.4     | 0.373                   |
| Entangled | Medium | Naïve  | Strict | 145            | 0.81      | 7.1     | 1.315                   |
| Entangled | Medium | Hybrid | None   | 146            | 0.76      | 8.6     | 0.096                   |
| Entangled | Medium | Hybrid | Strict | 144            | 0.79      | 7.7     | 0.388                   |
| Entangled | High   | Naïve  | None   | 404            | 1.00      | 0.1     | 0.273                   |
| Entangled | High   | Naïve  | Strict | 199            | 1.00      | 0.1     | 0.455                   |
| Entangled | High   | Hybrid | None   | 444            | 1.00      | 0.1     | 0.097                   |
| Entangled | High   | Hybrid | Strict | 168            | 1.00      | 0.2     | 0.313                   |

Values averaged over 10 seeds. Hybrid architecture reduces governance-induced price volatility by 57–82% across structured topologies (e.g., tree/high: naïve/strict  $\sigma = 1.15$  vs. hybrid/strict  $\sigma = 0.50$ ).

TABLE XI  
EXPERIMENT 6: MECHANISM ABLATION SUMMARY (NAÏVE  
ARCHITECTURE, AVERAGED OVER 10 SEEDS)

| Topology              | Load   | Random | EDF  | Greedy EV | Market |
|-----------------------|--------|--------|------|-----------|--------|
| <i>Welfare (a.u.)</i> |        |        |      |           |        |
| Tree                  | Medium | 22.1   | 22.1 | 23.0      | 23.0   |
| Tree                  | High   | 24.4   | 24.3 | 28.6      | 28.6   |
| SP                    | Medium | 17.8   | 17.7 | 17.4      | 17.4   |
| SP                    | High   | 2.4    | 2.5  | 5.7       | 5.7    |
| Entangled             | Medium | 0.3    | 0.3  | 8.4       | 8.4    |
| Entangled             | High   | 0.0    | 0.0  | 0.1       | 0.1    |
| <i>Drop rate (%)</i>  |        |        |      |           |        |
| Tree                  | Medium | 41.3   | 41.2 | 39.7      | 39.7   |
| Tree                  | High   | 53.5   | 53.7 | 48.8      | 48.8   |
| SP                    | Medium | 50.0   | 50.3 | 53.4      | 53.4   |
| SP                    | High   | 93.9   | 93.8 | 89.8      | 89.8   |
| Entangled             | Medium | 98.9   | 98.8 | 76.9      | 76.7   |
| Entangled             | High   | 100    | 100  | 99.9      | 99.8   |

TABLE XII  
THEORY-TO-EXPERIMENT MAPPING. EACH FORMAL RESULT IS PAIRED  
WITH ITS EMPIRICAL TEST AND KEY OBSERVABLES.

| Result                   | Experiment     | Observables                     |
|--------------------------|----------------|---------------------------------|
| Prop. 1 (Polymatroid)    | Exps. 1–2 (–S) | $\sigma_p$ , drop rate, scaling |
| Lemma 1 (GS valuations)  | Exp. 6 (–M)    | Welfare equivalence             |
| Prop. 2 (DSIC mechanism) | Exp. 6 (–M)    | Market $\approx$ value-greedy   |
| Prop. 3 (Encapsulation)  | Exp. 4 (–H)    | $\sigma_p$ reduction, latency   |

TABLE XIII  
COMPLETE ABLATION MATRIX

| Ablation                               | Source                       | Isolates                 |
|----------------------------------------|------------------------------|--------------------------|
| Full system (S+H+G+M)                  | Exp 5 (hybrid, strict)       | Everything on            |
| –M (remove market)                     | Exp 6 (greedy_ev vs. market) | Price coordination       |
| –G (remove governance)                 | Exp 5 (no gov conditions)    | Trust-gated capacity     |
| –H (remove hybrid)                     | Exp 4 (naïve conditions)     | Integrator encapsulation |
| –H <sub>eff</sub> (remove eff. factor) | Exp 4 (EMA only)             | Efficiency sub-component |
| –S (break structure)                   | Exp 1/4 (entangled)          | Structural discipline    |
| –M–H (heuristic + naïve)               | Exp 6 (EDF, naïve)           | Scheduling-only baseline |
| –S–H–M (worst case)                    | Exp 6 (random, entangled)    | Absolute floor           |